American International University- Bangladesh (AIUB)
Faculty of Engineering
Data Communications Lab

Course Name:	**Data Communications**		
Course Code:	CoE 3201	**Section:**	K
Semester:	Spring 2025-26	**Group No:**	05
Assignment Name:	**Open Ended Lab**		
Assessed CO5:	**Prepare and deliver effective technical presentations on experimental analysis of digital communication systems over noisy analog channels, clearly interpreting the impact of modulation order, bit rate, $Eb/N0$, and BER on system performance.**		
Assessed POI:	**P.j.3.A4**		
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Mark distribution (to be filled by Faculty):

Objectives	Indicator	Proficient [10-8]	Good [7-4]	Needs Improvement [3-1]	Secured Marks
Depth of knowledge displayed through appropriate research	P1	Student was able to apply in-depth engineering knowledge achieved by appropriate research about digital/analog communication to design the communication model correctly and fulfilled all design criteria.	Design process is not completely supported by in-depth engineering knowledge achieved by appropriate research about digital/analog communication, some but not all of the design criteria are fulfilled.	Design process contains mistakes and does not display enough in-depth engineering knowledge achieved by appropriate research about digital/analog communication. Most of the design criteria are not fulfilled.	
Depth of analysis	P3	Student defended the diversified approach taken to solve the problem with well-justified in-depth analysis demonstrated through necessary plots.	Student's attempts to analyze the diversified approach taken to solve the problem is not enough in-depth, some of analysis are not demonstrated through necessary plots.	Student did not attempt any in-depth analysis of the designed system and did not include most of the necessary plots.	
Level of integration of multiple sections of design for solution of high-level problem	P7	Student correctly identified all problems and successfully integrated the interdependent parts of the given problem.	Student was able to identify some of the problems correctly and integrated the interdependent parts of the given problem.	Student was able to identify only one/two of the problems correctly and could not properly integrate the interdependent parts of the given problem. Answer was incomplete.	
Comments:				**Total Marks (Out of 30):**	

		Total Marks (Converted to 20):	

Task

Question:

Suppose you want to send a message which contains the SURNAME NAME of any group member, all in capital letters. Using MATLAB, perform the following tasks and explain each step of your design.

1. **Convert the message to its binary ASCII Code.**
2. **Convert the ASCII code to the synchronous serial transmission format.**
3. **Convert digital data to a digital signal (unipolar NRZ signal with 1kbps data rate).**
4. **Now, modulate the digital signal using various digital modulation methods (BASK, 4-ASK, 16-ASK, QPSK, 8-PSK, 16-QAM) to send via a AWGN transmission channel and show the constellation diagrams.**
5. **Introduce some noise during transmission, assuming that the Eb/No of the channel is VAL1+20 dB and show the constellation diagrams (good case).**
6. **Demodulate the received signal using respective digital modulation.**
7. **Convert the binary data to retrieve the message.**
8. **Confirm if message in step 6 matches original message.**
9. **Change the Eb/No value and find the threshold of Eb/No where transmitted message cannot be reconstructed at the receiver anymore. Show the received signal constellation diagram (worse case) for this case.**
10. **Plot the bit error probility for Eb/No (10 to 30 dB) ranges for above mentioned modulation scemes.**

Parameters: Consider, your group number = **VAL1.**

Instructions:

1. Plagiarism is strictly prohibited.
2. Please use MATLAB software to accomplish the project.

Title: Performance Analysis of Digital Modulation Schemes over AWGN Channel using BER and Constellation Diagrams

Objective:

1. To analyze and compare the performance of different digital modulation schemes such as M-ASK, M-PSK, and M-QAM over an AWGN channel.
2. To evaluate the Bit Error Rate (BER) of each modulation technique at different values of Eb/N_0.
3. To observe and interpret constellation diagrams for different modulation schemes.
4. To understand the trade-off between data rate and error performance in M-ary modulation systems.
5. To determine which modulation technique performs better under noisy conditions.

Theory:

In digital communication systems, modulation techniques are used to transmit binary data efficiently over a communication channel. In this experiment, different M-ary modulation schemes such as M-ASK, M-PSK, and M-QAM are used to study their performance in the presence of noise. In M-ary systems, each symbol represents multiple bits specifically $\mathbf{log_2 M}$ bits per symbol which helps increase the data transmission rate.

M-ASK (M-ary Amplitude Shift Keying):

M-ASK is a modulation technique where the amplitude of the carrier signal is varied to represent different symbols. As the value of $\mathbf{M}$ increases more amplitude levels are introduced allowing more bits to be transmitted per symbol. However the distance between these amplitude levels becomes smaller and making the signal more sensitive to noise. Due to this even a small disturbance in the channel can cause errors at the receiver, resulting in a higher Bit Error Rate (BER) especially at lower $\mathbf{E_b/N_0}$values.

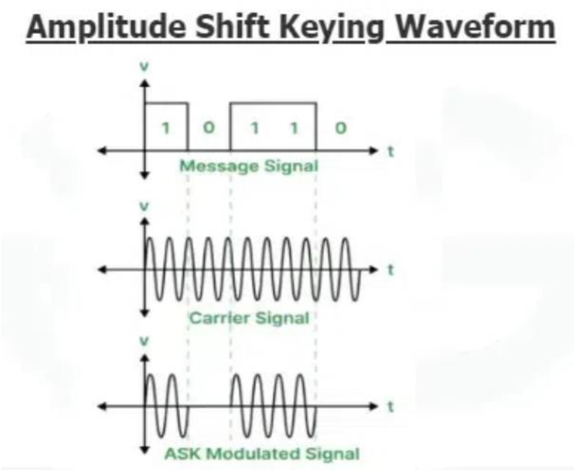


Fig: M-ASK

M-PSK (M-ary Phase Shift Keying) :

M-PSK works by changing the phase of the carrier signal instead of the amplitude. Each symbol is represented by a different phase angle. Since the amplitude remains constant so M-PSK is less affected by amplitude noise and generally performs better than M-ASK in noisy environments. However as the modulation order increases the phase difference between adjacent symbols becomes smaller which makes it more difficult for the receiver to correctly detect the signal. This leads to an increase in BER when the noise level is high.

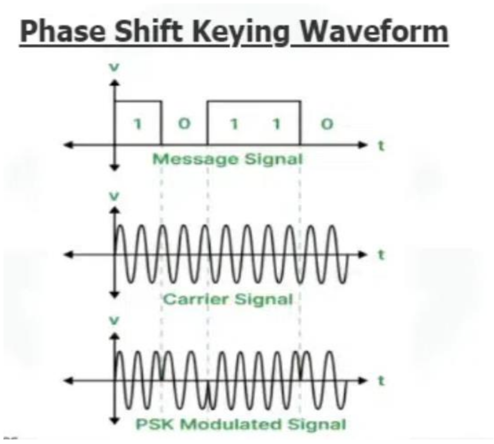


Fig:M-PSK

M-QAM (M-ary Quadrature Amplitude Modulation):

M-QAM combines both amplitude and phase variations to represent data. This allows it to transmit a higher number of bits per symbol making it more bandwidth-efficient compared to M-ASK and M-PSK. However because both amplitude and phase are affected by noise M-QAM becomes more sensitive in noisy conditions especially at higher modulation orders. The constellation points in M-QAM are closely spaced which increases the probability of symbol errors and thus leads to a higher BER at low $\mathbf{E_b/N_0}$.

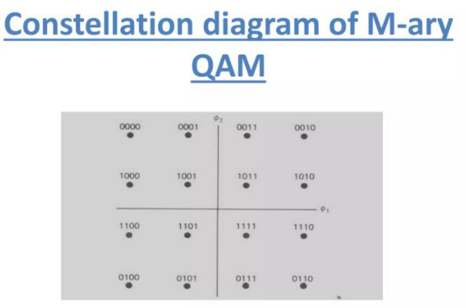


Fig: M-QAM

Bit Error Rate (BER):

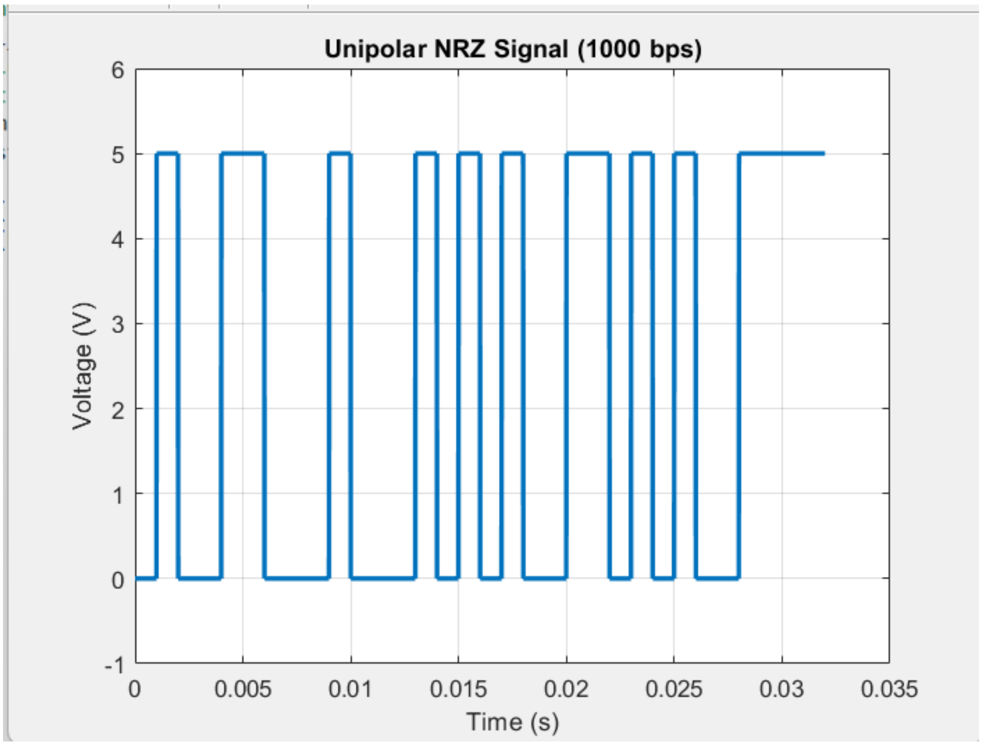
Bit error rate is a key parameter used to evaluate the performance of these modulation techniques. It represents the ratio of incorrectly received bits to the total transmitted bits. BER is strongly influenced by the value of $\mathbf{E_b/N_0}$ which indicates the signal strength relative to noise. A higher $\mathbf{E_b/N_0}$results in lower BER meaning better communication quality while a lower $\mathbf{E_b/N_0}$increases the chance of errors. From the analysis, lower-order modulation schemes provide better reliability with lower BER whereas higher-order schemes offer higher data rates but are more prone to errors. Therefore there is always a trade-off between data rate and reliability and the choice of modulation depends on the channel conditions.

Results:

Task: 1,2,3

```
clc; clear; close all;

% PARAMETERS & INPUT

VAL1 = 5; % Group Number
msg = 'LEMO';
disp(['Original Message: ' msg]);

%% TASK 1 & 2: MESSAGE -> BINARY -> SERIAL
% Convert to 8-bit binary ASCII
ascii_vals = double(msg);
% Using modern bit manipulation: de2bi is often replaced by int2bit in newer versions
% but for compatibility we use logical operations or bitget
bin_matrix = dec2bin(ascii_vals, 8) - '0';
bit_stream = reshape(bin_matrix', 1, []); % Serial stream

disp('Binary Data (First 16 bits):');
disp(bit_stream(1:16));

%% TASK 3: UNIPOLAR NRZ SIGNAL
bit_rate = 1000; % 1 kbps
samples_per_bit = 100;
t_bit = 1/bit_rate;
fs = samples_per_bit * bit_rate;
t = 0:1/fs:(length(bit_stream)*t_bit) - 1/fs;

% Generate Unipolar NRZ
digital_signal = zeros(1, length(t));
for i = 1:length(bit_stream)
    idx = (i-1)*samples_per_bit + 1 : i*samples_per_bit;
    digital_signal(idx) = bit_stream(i) * 5; % 5V for logic 1
end

figure('Name', 'Task 3: Digital Signal');
plot(t, digital_signal, 'LineWidth', 2);
ylim([-1 6]); grid on;
xlabel('Time (s)'); ylabel('Voltage (V)');
title(['Unipolar NRZ Signal (', num2str(bit_rate), ' bps)']);
```

Output:

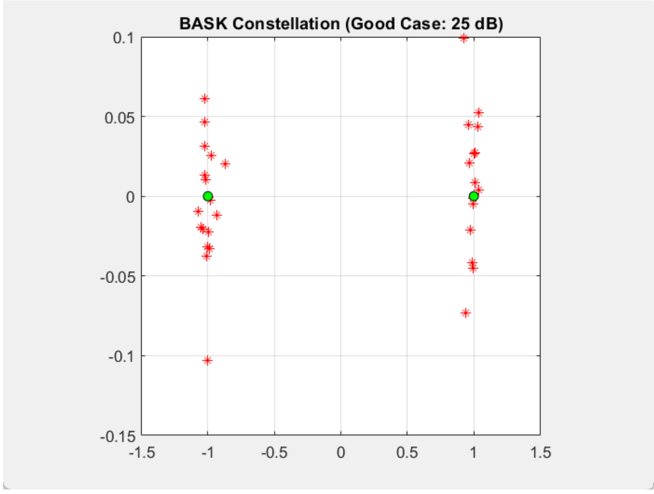


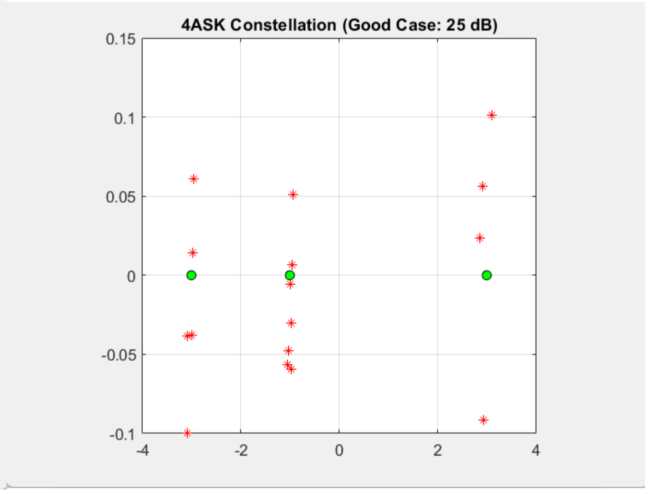
Task 4-8

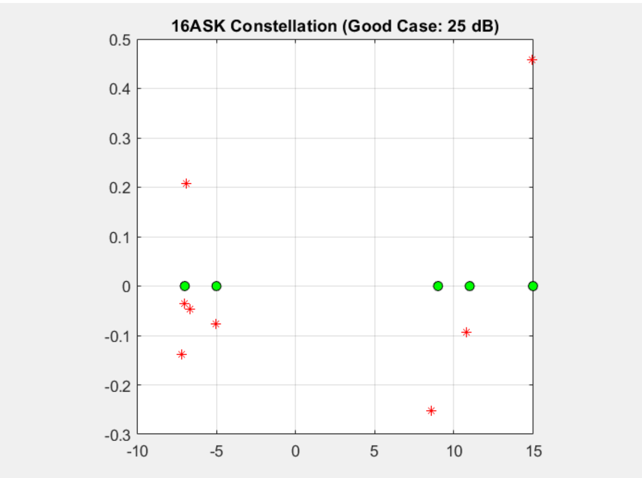
```matlab
%% TASK 4-8: MODULATION, NOISE, & RECOVERY
EbNo = VAL1 + 20; % 25 dB
schemes = {'BASK','4ASK','16ASK','QPSK','8PSK','16QAM'};
M_list = [2 4 16 4 8 16];

fprintf('\n--- Performance at Eb/No = %d dB ---\n', EbNo);

for i = 1:length(schemes)
    M = M_list(i);
    k = log2(M);

    % Padding bits to fit symbol size
    pad_len = mod(k - mod(length(bit_stream), k), k);
    padded_bits = [bit_stream, zeros(1, pad_len)];

    % Bit to Symbol mapping
    bits_per_sym = reshape(padded_bits, k, [])';
    symbols = bi2de(bits_per_sym, 'left-msb');

    % Modulation
    switch schemes{i}
        case {'BASK','4ASK','16ASK'}
            tx = pammod(symbols, M);
        case 'QPSK'
            tx = pskmod(symbols, M, pi/M); % Phase offset for QPSK
        case '8PSK'
            tx = pskmod(symbols, M);
        case '16QAM'
            tx = qammod(symbols, M, 'UnitAveragePower', true);
    end

    % AWGN Channel
    snr = EbNo + 10*log10(k); % Convert Eb/No to SNR
    rx = awgn(tx, snr, 'measured');

    % Demodulation
    switch schemes{i}
        case {'BASK','4ASK','16ASK'}
            demod_sym = pamdemod(rx, M);
        case 'QPSK'
            demod_sym = pskdemod(rx, M, pi/M);
        case '8PSK'
            demod_sym = pskdemod(rx, M);
        case '16QAM'
            demod_sym = qamdemod(rx, M, 'UnitAveragePower', true);
    end

    % Symbol to Bits
    rx_bits_padded = de2bi(demod_sym, k, 'left-msb')';
    rx_bits = rx_bits_padded(:)';
    rx_bits = rx_bits(1:length(bit_stream)); % Remove padding

    % Binary to String
    rx_chars = reshape(rx_bits, 8, [])';
    rx_msg = char(bi2de(rx_chars, 'left-msb'))';

    match_status = 'FAIL';
    if strcmp(msg, rx_msg), match_status = 'MATCHED'; end
    fprintf('Scheme: %-6s | Recovered: %-10s | Status: %s\n', schemes{i}, rx_msg, match_status);
```

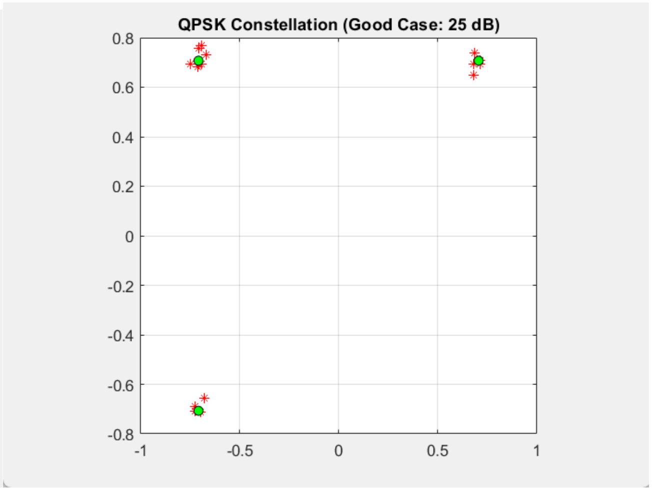
```
    % Plot Constellation (Task 5)
    figure('Name', ['Task 5: ' schemes{i}]);
    plot(real(rx), imag(rx), 'r*'); hold on;
    plot(real(tx), imag(tx), 'ko', 'MarkerFaceColor', 'g');
    title([schemes{i} ' Constellation (Good Case: ' num2str(EbNo) ' dB)']);
    grid on; axis square;
end
```

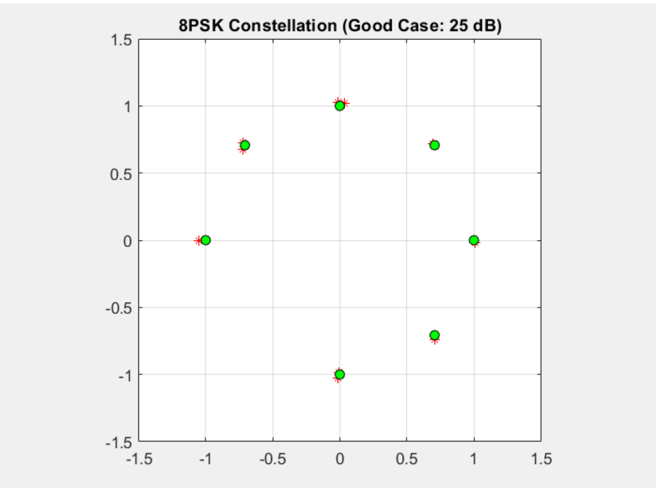
Output:

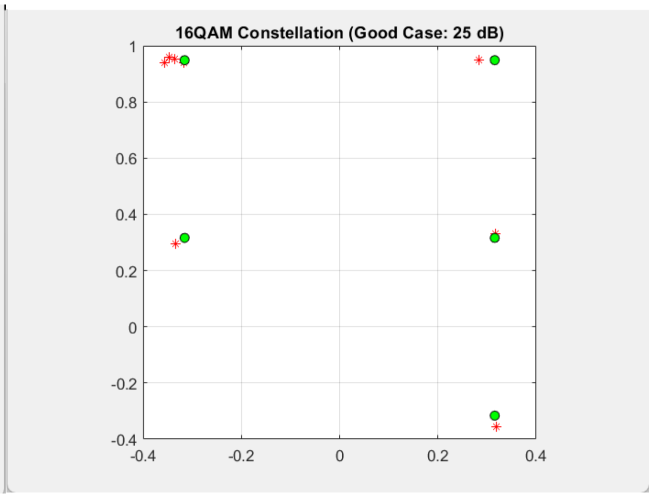












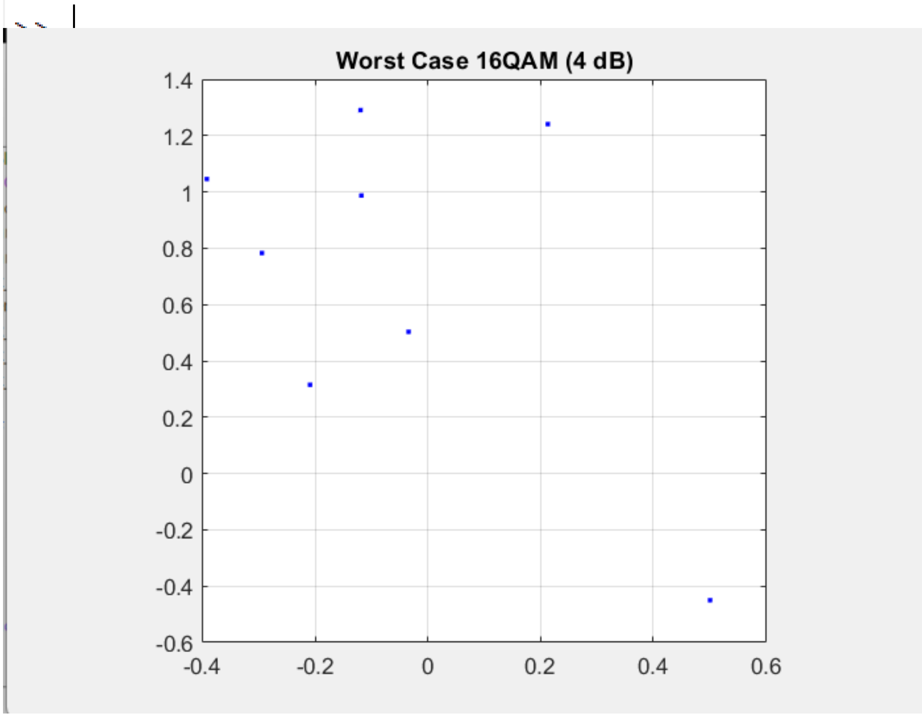
```
Original Message: LEMO
Binary Data (First 16 bits):
     0     1     0     0     1     1     0     0     0     1     0     0     0     1     0     1


--- Performance at Eb/No = 25 dB ---
Scheme: BASK   | Recovered: LEMO       | Status: MATCHED
Scheme: 4ASK   | Recovered: LEMO       | Status: MATCHED
Scheme: 16ASK  | Recovered: LEMO       | Status: MATCHED
Scheme: QPSK   | Recovered: LEMO       | Status: MATCHED
Scheme: 8PSK   | Recovered: LEMO       | Status: MATCHED
Scheme: 16QAM  | Recovered: LEMO       | Status: MATCHED
```

Task 9

```matlab
%% TASK 9: WORST CASE THRESHOLD (16-QAM)
disp('Checking threshold for 16QAM...');
threshold_found = false;
for current_ebno = 20:-1:-5 % Stepping down
    snr = current_ebno + 10*log10(log2(16));
    rx_noise = awgn(tx, snr, 'measured'); % Using tx from last loop (16QAM)
    demod_test = qamdemod(rx_noise, 16, 'UnitAveragePower', true);
    rx_bits_test = de2bi(demod_test, 4, 'left-msb')';
    rx_bits_test = rx_bits_test(:)';
    rx_msg_test = char(bi2de(reshape(rx_bits_test(1:length(bit_stream)), 8, [])', 'left-msb'))';

    if ~strcmp(msg, rx_msg_test)
        fprintf('Worst Case Found! Eb/No = %d dB failed.\n', current_ebno);
        figure('Name', 'Task 9: Worst Case');
        plot(real(rx_noise), imag(rx_noise), 'b.');
        title(['Worst Case 16QAM (' num2str(current_ebno) ' dB)']);
        grid on; axis square;
        threshold_found = true;
        break;
    end
end
```

Output:

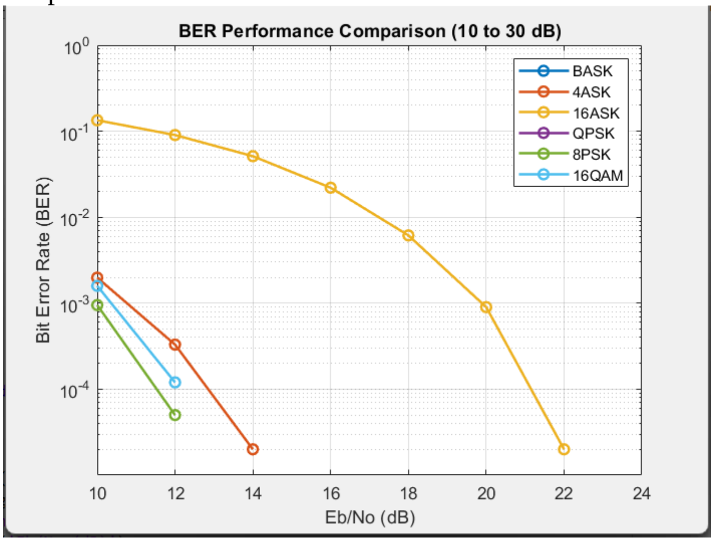
```
Checking threshold for 16QAM...
Worst Case Found! Eb/No = 4 dB failed.
>>
```



Task 10:

```
%% TASK 10: BER vs Eb/No

EbNo_range = 10:2:30;
ber_results = zeros(length(schemes), length(EbNo_range));
num_bits = 10^5; % Sufficient bits to catch errors at 10-15dB

for i = 1:length(schemes)
    M = M_list(i);
    k = log2(M);

    % Generate random bits for statistical accuracy
    test_bits = randi([0 1], 1, num_bits - mod(num_bits, k));
    tx_bits_mat = reshape(test_bits, k, [])';
    tx_sym = bi2de(tx_bits_mat, 'left-msb');

    % Modulate based on scheme
    switch schemes{i}
        case {'BASK','4ASK','16ASK'}, sig = pammod(tx_sym, M);
        case 'QPSK', sig = pskmod(tx_sym, M, pi/M);
        case '8PSK', sig = pskmod(tx_sym, M);
        case '16QAM', sig = qammod(tx_sym, M, 'UnitAveragePower', true);
    end

    for j = 1:length(EbNo_range)
        % Convert Eb/No to SNR for the channel
        current_snr = EbNo_range(j) + 10*log10(k);

        % Add Noise
        rx_sig = awgn(sig, current_snr, 'measured');

        % Demodulate
        switch schemes{i}
            case {'BASK','4ASK','16ASK'}, d_sym = pamdemod(rx_sig, M);
            case 'QPSK', d_sym = pskdemod(rx_sig, M, pi/M);
            case '8PSK', d_sym = pskdemod(rx_sig, M);
            case '16QAM', d_sym = qamdemod(rx_sig, M, 'UnitAveragePower', true);
        end

        % Calculate BER
        rx_bits_mat = de2bi(d_sym, k, 'left-msb')';
        [~, ber_results(i,j)] = biterr(test_bits, rx_bits_mat(:)');
    end
end

% PLOT THE DIAGRAM
figure('Name', 'Task 10: BER Curves (10-30 dB)');
semilogy(EbNo_range, ber_results, '-o', 'LineWidth', 1.5);
grid on;
xlabel('Eb/No (dB)');
ylabel('Bit Error Rate (BER)');
legend(schemes);
title('BER Performance Comparison (10 to 30 dB)');
ylim([1e-5 1]); % Standard log scale for BER
```

Output:



Result analysis:

Task 1&2: ASCII and Serial Data Conversion

The input message was first converted into its binary ASCII representation and then arranged into a serial bit stream. This step ensures that the data is properly formatted for transmission. The output confirms that each character is correctly mapped into binary form, forming the foundation of the communication system.

Task 3: Unipolar NRZ Signal Generation

The binary data was converted into a unipolar NRZ signal where '1' is represented by a high voltage and '0' by zero voltage. The output waveform clearly shows this pattern, confirming correct digital signal generation at the given bit rate (1 kbps). This stage determines the basic bit rate of the system.

Task 4 & 5: Modulation and Constellation Diagrams (Without and with noise)

Different modulation techniques (BASK, M-ASK, QPSK, M-PSK, and M-QAM) were applied to the signal. The constellation diagrams show clearly separated points without any distortion. Lower-order modulation schemes have fewer points with larger (spacing), while higher-order schemes (like 16-QAM) have more points placed closer together. This shows that increasing modulation order increases the number of bits per symbol (higher bit rate), but reduces the distance between symbols, making them more sensitive to noise. We use a single constellation diagram for each scheme to plot the ideal and with noise (good case) bits to make the visualization easier. The green dots represent the ideal(without noise) state and the red asterisks represent the noisy state. When noise was introduced using a high E_b/N_0, the constellation points became slightly scattered but still remained distinguishable. This indicates that the signal power is strong compared to noise. Since E_b/N_0 is high, the Signal-to-Noise Ratio (SNR) is also high, resulting in minimal bit errors. The system still performs reliably under this condition.

Task 6: Demodulation of Received Signal

The received noisy signals were demodulated using their respective techniques. Since the constellation points were still clearly grouped (in good conditions), the receiver could correctly identify the transmitted symbols. This shows that proper demodulation is possible when noise levels are manageable.

Task 7&8: Message Reconstruction and Verification

After demodulation, the binary data was converted back into the original message. The output confirms that the transmitted and received messages match under good channel conditions. This verifies the correctness of the overall communication system.

Task 9: Effect of Low E_b/N_0 (Worst Case Condition)

When E_b/N_0 was reduced to 4 dB, the 16-QAM constellation points became highly scattered and lost their structural integrity. The noise power is now comparable to the signal power, causing symbols to shift across decision boundaries. Consequently, the receiver cannot correctly distinguish between symbols, leading to decoding errors and the failure to recover the original message. This demonstrates that lower E_b/N_0 significantly increases BER and reduces system reliability.

Task 10: BER vs E_b/N_0 Analysis

The BER graph shows that BER decreases as E_b/N_0 increases for all modulation schemes. At low E_b/N_0, BER is high due to strong noise influence. As E_b/N_0 increases, the signal becomes stronger relative to noise, reducing errors. The effect of modulation order is also clear here:

- Lower-order modulation (BASK, QPSK): lower BER, better reliability
- Higher-order modulation (16-ASK, 16-QAM): higher BER at low E_b/N_0, but higher bit rate

Higher-order modulation transmits more bits per symbol, increasing data rate, but requires higher energy per bit to maintain performance. This leads to increased energy consumption in the system.

Conclusion:

This experiment focused on analyzing the performance of different digital modulation schemes over an AWGN channel using BER and constellation diagrams. The results demonstrate that as the modulation order increases, the constellation points become more closely spaced, making the system more vulnerable to noise. This effect was clearly visible in the constellation diagrams, where higher-order schemes showed greater symbol scattering under noisy conditions. The BER analysis further supports this observation, showing that lower-order modulation techniques such as BASK and QPSK achieve lower error rates, while higher-order schemes like 16-QAM experience higher BER, especially at low E_b/N_0 values. However, higher-order modulation provides increased data rates, indicating a trade-off between efficiency and reliability. Overall, the experiment confirms that system performance strongly depends on both the modulation technique and channel conditions, and appropriate selection is necessary to balance data rate and error performance.

Reference:

[1] J. G. Proakis and M. Salehi, Fundamentals of Communication Systems, 2nd ed. Upper Saddle River, NJ, USA: Pearson Education, 2014.

[2] B. P. Lathi and Z. Ding, Modern Digital and Analog Communication Systems, 5th ed. Oxford, UK: Oxford Univ. Press, 2018.

[3] The MathWorks, Inc., "MATLAB Documentation," Natick, MA, USA. [Online]. Available: https://www.mathworks.com/help

[4] A. V. Oppenheim, A. S. Willsky, and S. H. Nawab, Signals and Systems, 2nd ed. Upper Saddle River, NJ, USA: Prentice Hall, 1997.

[5] S. Haykin and M. Moher, Introduction to Analog and Digital Communications, 2nd ed. Hoboken, NJ, USA: John Wiley & Sons, 2007.